

Stability Analysis of Closed-Loop DC Motor

3.1 Aim

Evaluate the performance of the closed-loop DC motor by testing various gain values and assessing the corresponding Gain and Phase Margins values.

3.2 List of Equipment

Table 3.1: List of Equipment

S.No.	Equipment Name	Manufacturer
1	STM32 Microcontroller	F767zi
2	Encoder Motor	1
3	Breadboard	1
4	Motor Driver	1

3.3 Theory

The stability of a closed-loop control system can be analyzed using frequency domain methods such as Gain Margin (GM) and Phase Margin (PM).

- **Gain Margin (GM):** The amount by which the system gain can be increased before the system becomes unstable. It is found at the frequency where the phase of the system is -180° .
- **Phase Margin (PM):** The extra phase angle that can be added before the system becomes unstable. It is found at the frequency where the system gain is equal to 1 (0 dB).

Oscillatory Nature of System Response

- **Underdamped system ($0 < \zeta < 1$):** The system output oscillates around the final value with gradually decreasing amplitude. This response is faster but includes overshoot before settling.
- **Critically damped system ($\zeta = 1$):** The system reaches the steady-state value in the shortest possible time without oscillations. It gives the fastest non-oscillatory response.
- **Overdamped system ($\zeta > 1$):** The system responds very slowly and takes a longer time to reach the final value. There are no oscillations, but the response is sluggish.

The damping factor (ζ) controls whether the system will oscillate or not, while the time constant (τ) determines how fast the response settles. Together, ζ and τ define the stability and oscillatory behavior of the closed-loop DC motor system.

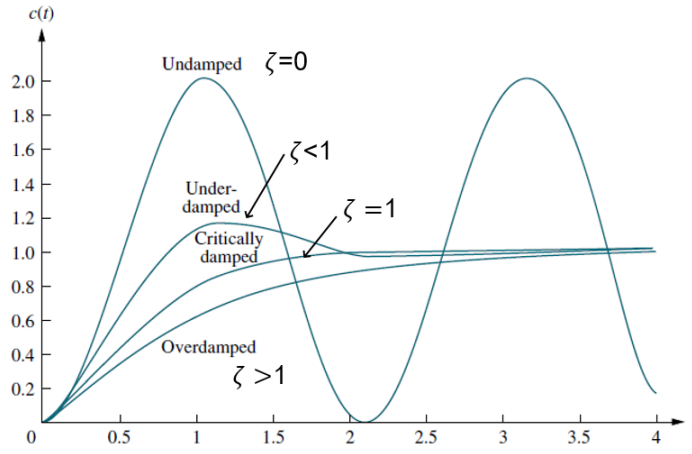


Figure 3.1: Damping Torque Definition and Types

Procedure

1. Connect the DC motor setup with the STM32 board.
2. Perform system identification to obtain the motor transfer function.
3. Replace the identified motor transfer function in the Simulink block diagram.
4. Construct a closed-loop system with a proportional gain block.

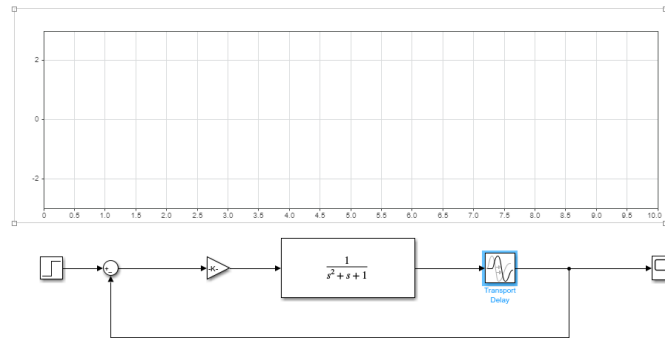


Figure 3.2: Step 11: Simulink Block Diagram

5. Vary the gain values and observe the system response.

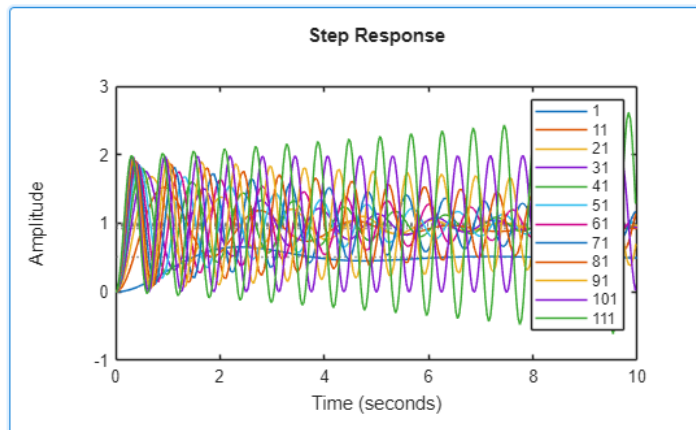


Figure 3.3: Step Response at various gain margin

Step response at various gain margin

6. Identify the condition of sustained oscillations (sinusoidal steady-state output). Step Response (Sustained oscillations)
7. Note the Gain Margin (GM) and Phase Margin (PM) values for different gain settings.
8. Record the critical gain value at which sustained oscillations occur.
9. Plot the graph of Gain vs. Gain Margin.

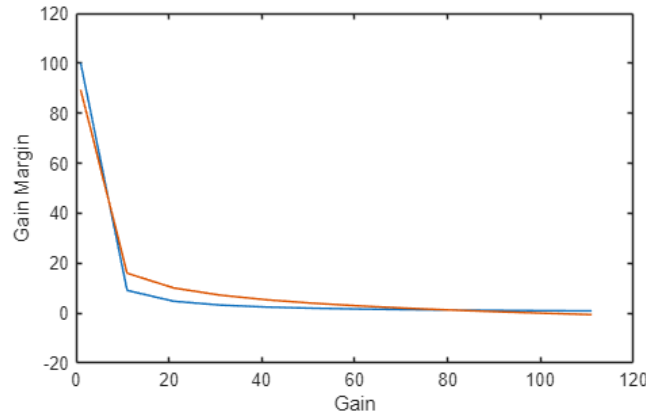


Figure 3.4: Gain Vs Gain Margin

Observation

S.No	Gain Value (K)	Gain Margin (dB)	Phase Margin (deg)
1			
2			
3			
4			
5			
8			

Table 3.2: Measured Gain Margin and Phase Margin at different gain values

Result and Conclusion

The transfer function of the DC motor was obtained using STM32-based system identification, and the closed-loop response was analyzed for different gain values. The critical gain value corresponding to sustained oscillations was identified, and Gain Margin (GM) and Phase Margin (PM) were determined for each operating condition. It was observed that the system response and stability depend on the damping factor (ζ) and the time constant (τ), which define whether the system is underdamped, critically damped, or overdamped. Sustained oscillations at the critical gain verify the stability limits of the closed-loop DC motor system.